Simulation Results

Sensitivity Analysis: Ramp Count vs Accessibility Outcomes

1. Model Setup

The model simulates pedestrian movement along an urban footpath corridor with a single kerb crossing. Three agent types move eastward: general pedestrians (1.2 m/s), wheelchair users (0.5 m/s), and elderly pedestrians (0.8 m/s). A fixed random seed (random-seed 42) was added to the setup procedure. This ensures the obstacle layout is identical across all five runs, so ramp count is the only variable affecting results.

2. Fixed Parameters

Parameter	Value
obstacle-density	0.20
ramp-search-radius	8 patches
num-general	15 agents
num-wheelchair	8 agents
num-elderly	8 agents
Random seed	42 (ensures identical obstacle layout across all runs)
Max simulation duration	2000 ticks

3. Results

ramp-count	Total blockages	% blocked	Avg time — wheelchair	Avg time — general	Accessibility gap
0	**156**	**100%**	140.0 ticks	59.5 ticks	80.5 ticks
5	0	0%	132.5 ticks	59.5 ticks	73.0 ticks
10	0	0%	135.9 ticks	59.5 ticks	76.3 ticks
15	0	0%	128.0 ticks	59.5 ticks	68.5 ticks
20	0	0%	131.8 ticks	59.5 ticks	72.2 ticks

General pedestrian time is 59.5 ticks across all runs — consistent because random-seed 42 fixes the obstacle layout.

4. Three-Zone Pattern

Zone	ramp-count	What happens
Exclusion	0	156 blockages, 100% of wheelchair users blocked, many do not complete the corridor
Transition	5	Zero blockages, all wheelchair users complete the journey, accessibility gap 73.0 ticks
Saturation	10 – 20	No further improvement, travel time stays between 128.0 and 135.9 ticks

5. Findings

5.1 Non-linear blockage threshold

The sharpest result is the transition between ramp-count = 0 and ramp-count = 5. At zero ramps, 156 blockage events were recorded and every wheelchair user was blocked — 100%. At five ramps, blockages dropped to zero. This complete elimination happened in one step, not gradually. The threshold was not programmed into any agent rule — it emerged from the spatial interaction between agent movement paths and ramp positions along the kerb column across many simulation ticks.

5.2 Persistent accessibility gap

Even after blockages are eliminated at ramp-count = 5, wheelchair users consistently take between 68.5 and 76.3 ticks longer than general pedestrians to complete the same corridor. The accessibility gap does not close with ramp provision alone. It reflects the underlying speed difference (wheelchair users: 0.5 m/s vs general pedestrians: 1.2 m/s) combined with the detour cost of navigating to ramp locations.

5.3 Saturation point

From ramp-count = 5 onwards, wheelchair travel time stays between 128.0 and 135.9 ticks regardless of whether 5, 10, 15, or 20 ramps are present — a variation of less than 6%. Adding ramps beyond 5 produces no meaningful improvement in travel time. The bottleneck is no longer ramp availability but agent walking speed and the obstacle layout on the post-kerb section of the corridor.

The three-zone pattern is a clean result of ramp count alone because random-seed 42 controls for stochastic obstacle variation.

6. Policy Implications

1. **Zero-ramp streets are fully exclusionary.** At ramp-count = 0, not a single wheelchair user completed the corridor. 156 blockage events. This is inconsistent with the RPwD Act (2016).

2. **Minimum viable threshold is 15% kerb coverage.** Five ramps on a 33-patch kerb eliminated all blockages. Targeted investment above this threshold is more effective than sparse uniform provision.
3. **Ramp provision alone is insufficient for equitable mobility.** The accessibility gap of 68.5 to 76.3 ticks requires complementary measures beyond ramp installation.
4. **Saturation point limits marginal returns.** Beyond 5 ramps, additional provision produces less than 6% variation in travel time. Infrastructure budgets should prioritise reaching the threshold on more streets.

7. Limitations

- **Single run per configuration** — multiple runs with averaged outputs would be more statistically rigorous
- **Grid geometry** — does not capture real footpath irregularity, gradient, or surface texture
- **Literature-derived speeds** — not calibrated from field observation of Indian pedestrians
- **Agent homogeneity within types** — all wheelchair users share identical parameters
- **No social dynamics** — yielding, assistance, and social navigation absent